IMADPort v1.0.1

Interval Mean Absolute Deviation Portfolios v1.0.1

User Manual

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Preface

This manual provides reference information about the IMADPort software version 1.0.1 distributed for free through the GitHub repository, targeting a user who has little experience in using a command line interface (CLI). A basic knowledge of the interval portfolio theory is also included so that a user can fully understand the main purpose of the software development. As the initial version, the program is built dynamically and lacks a graphical user interface (GUI) to hightlight its mere functionality and, more importantly, to have small file size. MATLAB runtime (MCR)—a set of shared libraries—must be loaded into memory at the start of the program, leading to the slightly slow initial load time of the program. Furthermore, the CLI deployment provides more flexibility than the sole GUI; it facilitates through scripting the automation of obtaining bounds on optimal portfolios of different types.

Contents

1 Interval Portfolio Theory 1

2 Installation 2

3 Usage 4

References 7

1 Interval Portfolio Theory

A financial asset not limited to a stock has a change in its value over time and can generate either a loss or a profit. Since its future, usually one-period ahead, price is not perfectly predictable, the investment risk which measures the deviation, using its historical asset returns dating back time periods, of its realized return from its expected value plays an important role in asset selection. A portfolio including multiple assets of different characteristics aims to mitigate this particular risk.

In addition to a future asset return, its expected value is also uncertain. This is attributable to the existence of several good estimates beyond the arithmetic mean of past asset returns. All of these candidates may form a closed interval, optimal portfolios exist, and asset allocation is suggested through bounds, not single values as usual. Intuitively, a good investment strategy should guarantee a minimum rate of return specified by an investor. This becomes a portfolio constraint. An optimal portfolio usually attains a return more than this threshold.

A portfolio risk should also be minimized because an investor prefers an actual return as close to an expected return as possible. Any value of this magnitude is equally important; therefore, the mean absolute deviation (MAD) should be employed as a risk measure (see Konno & Yamazaki, 1991, for the first proposal of the MAD portfolio model) (see Feinstein & Thapa, 1993; Speranza, 1994, for subsequent model developments). The rigorous derivation of bounds on optimal MAD portfolios (Chaiyakan & Thipwiwatpotjana, 2019, 2021) heavily relies on mathematical tools such as the foundation of interval linear programming (see Fiedler, Nedoma, Ramík, Rohn, & Zimmermann, 2006; Hladík, 2009, 2012, for example) and the concept of basis stability (Hladík, 2014; Rohn, 1993).

2 Installation

The IMADPort software can run on all three 64-bit operating systems: Windows (AMD64 or x64), macOS (Apple Silicon), and Linux (AMD64 or x64). It requires the following components of the MATLAB runtime (MCR) R2026a.

- MATLAB Base Runtime
- MATLAB Standard Runtime Addon
- MATLAB Graphics Runtime Addon
- Optimization Toolbox Runtime Addon

Developed by MathWorks, MATLAB R2026a, if preinstalled, comes with this runtime. Unlike the MATLAB software, the MCR itself is available for free at https://www.mathworks.com/products/compiler/matlab-runtime.html. Nonetheless, the MCR under this approach is considerably large and has a size of at least 10 GB because all components beyond requirements are installed. To save disk space, the minimal online MCR installer is also separately provided by the developer of the IMADPort software. Through this installation method, only four MCR dependencies above with a total size of approximately 2 GB are fetched from MathWorks servers and installed.

As shown in the table below, the IMADPort directory and the MCR directories should be prepended to the PATH and library path environment variables, respectively. This setup varies based on an operating system. Paths are separated by a semicolon on Windows but a colon on both macOS and Linux.

Operating system	Environment variables and directories
Windows	PATH <*IMADPORT_INSTALL_DIR*> <*MATLAB_RUNTIME_INSTALL_DIR*>\runtime\win64
macOS	PATH <*IMADPORT_INSTALL_DIR*>/imadp.app/Contents/MacOS
	DYLD_LIBRARY_PATH <*MATLAB_RUNTIME_INSTALL_DIR*>/runtime/maca64 <*MATLAB_RUNTIME_INSTALL_DIR*>/bin/maca64 <*MATLAB_RUNTIME_INSTALL_DIR*>/sys/os/maca64 <*MATLAB_RUNTIME_INSTALL_DIR*>/extern/bin/maca64
Linux	PATH <*IMADPORT_INSTALL_DIR*>
	LD_LIBRARY_PATH <*MATLAB_RUNTIME_INSTALL_DIR*>/runtime/glnxa64 <*MATLAB_RUNTIME_INSTALL_DIR*>/bin/glnxa64 <*MATLAB_RUNTIME_INSTALL_DIR*>/sys/os/glnxa64 <*MATLAB_RUNTIME_INSTALL_DIR*>/extern/bin/glnxa64

3 Usage

Through the single imadp binary executable, the IMADPort software is available from a command line interface (CLI) only, without a graphical user interface (GUI). As interpreters, the default command line shells are both Command Prompt (CMD) and PowerShell on Windows, Z shell (Zsh) on macOS, and Bourne-again shell (Bash) on Linux. Providing text-based interfaces to interact with shells, the default terminal emulators are both Windows Console Host (conhost.exe) and Windows Terminal on Windows, Terminal (Terminal.app) on macOS, and GNOME Terminal on Linux.

The --help flag can be used to display the usage information including a list of arguments. The imadp execution without flag also displays this help message. Only a CSV input file path is required and must be assigned to the first argument. Unless specified, a portfolio threshold return is 0 by default indicating a no-loss investment. The software can determine the feasibility of a threshold input and, if possible, suggest additional tighter bounds on optimal asset allocation. The optimal asset weights can be exported for future use. In this case, the CSV output file path must be supplied to the last argument, regardless of whether its directory exists. The calculated bounds on optimal portfolio risks and attainable portfolio returns are also displayed on a terminal emulator and can be saved by redirecting stdout (standard output) to a text file.

```
Interval MAD Portfolios v1.0.1
Developed by Songkomkrit Chaiyakan, Thaksin University
MATLAB runtime R2026a required

Help Message
All returns must be expressed in numerical ratios (not in percentage)
Argument 1: Input file in CSV format with header included (required)
    Column 1: Asset symbol
    Column 2: Lower bound on expected asset returns
    Column 3: Upper bound on expected asset returns
    Column 4 onward: Historical asset returns
Argument 2: Portfolio threshold return (optional) (default: 0)
Argument 3: Output of optimal asset weights in CSV format (optional) (default: None)
```

To illustrate, an MAD portfolio of four stock market indexes is considered. The historical market returns across eight time periods and the bounds on the expected future market returns are provided in the table below.

Symbol	LB	UB	R1	R2	R3	R4	R5	R6	R7	R8
Index 1	1.0251	1.0264	1.0043	0.9306	1.0179	0.9082	1.0787	1.0297	1.0179	1.0393
Index 2	1.0165	1.0181	1.0105	0.9491	0.9793	0.9639	1.0358	1.0152	1.0289	1.0191
Index 3	1.0334	1.0344	0.9964	0.8989	1.0611	0.9751	1.0811	1.0247	1.0146	1.0223
Index 4	1.0118	1.0136	1.0549	0.9088	1.0196	0.8955	1.0379	1.0294	0.9916	1.0497

Suppose the CSV input and, if specified, output files are located at <*INPUT_DIR*>/index.csv and <*OUTPUT_DIR*>/pindex.csv, respectively. If an investor wants to generate a portfolio return of at least 0 which is also a default setting in the IMADPort software, all of the following four commands produce the same set of optimal portfolios.

- imadp <*INPUT_DIR*>/index.csv
- imadp <*INPUT_DIR*>/index.csv 0
- imadp <*INPUT_DIR*>/index.csv <*OUTPUT_DIR*>/pindex.csv
- imadp <*INPUT_DIR*>/index.csv 0 <*OUTPUT_DIR*>/pindex.csv

Nonetheless, only the last two commands automatically export the optimal asset allocation in CSV file format in addition to displaying the result on the screen as shown below. The normal bounds on optimal asset weights are in the Lower and Upper columns, whereas the improved bounds, if existing, in the ImpLower and ImpUpper columns.

```
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Developed by Songkomkrit Chaiyakan, Thaksin University
MATLAB runtime R2026a required

Inputs
Portfolio threshold: 0.000000
Export optimal asset weights: true

Optimal Portfolios
Recommended threshold returns: between 1.011800 and 1.033400
Risks: between 0.024823 and 0.025184
Attainable returns: between 1.015884 and 1.017970
Improved attainable returns: between 1.016173 and 1.017400

Optimal Asset Weights
Symbol      Lower       Upper       ImpLower    ImpUpper
Index 1     0.000000    0.029489    0.000000    0.000000
Index 2     0.787357    1.000000    0.844469    0.930484
Index 3     0.000000    0.103097    0.000000    0.000000
Index 4     0.000000    0.155531    0.069516    0.155531
```

The CSV output file *<OUTPUT_DIR>*/pindex.csv is created at the request of a program user.

```
Symbol,Lower,Upper,ImpLower,ImpUpper
Index 1,0,0.029489,0,0
Index 2,0.78736,1,0.84447,0.93048
Index 3,0,0.1031,0,0
Index 4,0,0.15553,0.069516,0.15553
```

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